

From Wearable Data to Risk Literacy: Visualizing Physiological Heat Stress for Rickshaw Pullers in Dhaka

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Abstract

Visual tools for communicating physiological risk to low-literacy informal workers remain scarce. We present a mixed-method study in which 11 rickshaw pullers were shown aggregated visualizations of wearable data: one weather marker and three biomarkers derived from a three-year-long dataset of 100 rickshaw pullers. Using a pre and post-risk assessment protocol paired with semi-structured interviews, we examine how participants perceive and respond. We find that visualizations were associated with an upward shift in reported risk ratings across all biomarkers, with relative cardiac cost (RCC) showing the largest observed shift, though these shifts did not reach statistical significance at our sample size. Subsequently, thematic analysis reveals four themes: informal health knowledge, risks beyond biomarkers, individual recovery strategy, and normalization. Based on this analysis, we propose design considerations for just-in-time and post-hoc intervention, as well as implications for wearable computing and interface design for low-literacy populations.

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1 Introduction

Wearable Personal informatics devices for managing occupational heat stress remain poorly suited to frontline outdoor workers. Existing wearable designs (Apple Watch, Fitbit, Oura Ring, etc.) largely target white-collar users, producing a mismatch between current design assumptions and the realities of field work that demands sustained, intensive physical exertion [4]. Rickshaw pulling in Dhaka, Bangladesh, exemplifies this gap: pullers perform heavy labor outdoors across seasons of extreme heat and humidity, yet have little access to information about the physiological toll of that exposure.

Prior work has proposed a Bayesian Belief Network for predicting key physiological biomarkers of rickshaw pullers from demographic, activity, and weather data, coupled with climate-model forecasts to project these biomarkers under future climate scenarios for Dhaka [11]. The resulting wearable sensing dataset includes 100 pullers monitored across winter, summer, and monsoon conditions from 2023 to 2025. Notably, rickshaw pullers in prior work have expressed curiosity about the meaning and implications of their physiological measurements, highlighting an information gap.



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The present study investigates this gap by asking the following questions.

- (1) RQ1: Does exposure to aggregated weather and physiological biomarker visualizations (as annotated bar charts) shift rickshaw pullers' perception of heat-related risk?
- (2) RQ2: To what extent can rickshaw pullers who represent low literacy groups in society interpret and engage with such scientifically grounded visualizations?
- (3) RQ3: How do rickshaw pullers describe their adaptation and recovery strategies in relation to heat-related risk?

In exploring these research questions, we find some relevant studies which we summarize in Table 1.

Liu et al. [8] studied a general USA-based population's perception of binary climate data visualizations spanning eight decades of historical data. Unlike the studies below, this work presented aggregated data, surfacing design considerations for how non-expert audiences interpret long-term climate trends. Solano-Kamaiko et al. [13] gave home care workers personalized activity-tracking visualizations of their own data. Reflecting on personal data encouraged many participants and motivated behavior change. Okolo et al. [9] examined how community health workers in rural India perceive explainable AI interfaces, concluding that standard XAI methods are poorly suited to low-literacy, novice users in the Global South and calling for visualizations that avoid color associations, complex numerical labels, and multi-image layouts. Most directly related, Chaudhary et al. [4] deployed personal informatics sensors with strawberry farmworkers in California, presenting each worker's own physiological and work statistics back to them. Despite initial enthusiasm, workers normalized their grit, prioritizing earnings over health.

Taken together (Table 1), these studies inform design in three ways. First, tools designed for general or higher-literacy users do not necessarily transfer to populations with minimal formal education, and even within frontline contexts their usefulness varies [4, 13]. Second, the temporal window of the data shapes how workers trust and interpret it. Third, presenting individuals with their own longitudinal data risks memory and outlier bias, where anomalous readings distort perception. We therefore present (i) bar chart-based binary visualizations for ease of understanding and (ii) aggregate data drawn from three years of physiological and weather measurements across the broader rickshaw puller cohort ($n=100$).

2 Method

We conducted a semi-structured interview study in which rickshaw pullers in Dhaka were shown visual summaries of physiological and weather data drawn from an earlier cohort of 100 pullers [11], and rated their perceived heat-related risk before and after exposure to those visuals (Figure 1).

2.1 Visualization Design

We prepared five visual stimuli (Figure 2). The first served as an introduction to the study protocol: a rest-ride-rest schematic (Figure 2a) illustrating the three-phase data collection procedure: 5–10 minutes of rest, a 15–20 minute rickshaw ride covering 1.6 km, and a second rest period that each participant in the original cohort had undergone [11]. Next, weather context (wet bulb globe temperature, or “perceived temperature”) and three others for each

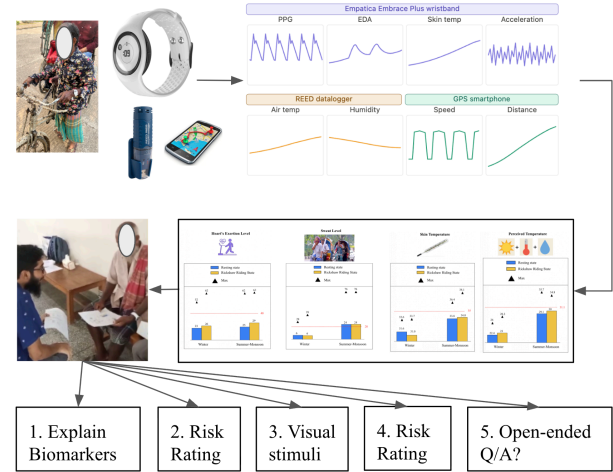


Figure 1: Study protocol overview.

biomarker: skin temperature; relative cardiac cost, or “degree of the heart’s exertion”; and skin conductance level, or “sweat level”. Each visualization is a grouped bar chart contrasting two seasons, winter versus summer–monsoon, and within each season two activity states, resting and driving, with a triangular marker denoting the maximum observed value. Our visualizations are inspired by the fact that a binary visual approach helps to better understand the perceived impact [8].

The red threshold lines in each visualization correspond to occupationally established risk boundaries. For WBGT, we apply the established $>31.1^{\circ}\text{C}$ limit as “High risk of heat-related illness or Extreme conditions” [3, 12, 14]. For skin temperature, we use 35°C , the upper limit of normal thermoregulatory range, beyond which sustained elevation risks driving core temperature to lethal values [12, 15]. For relative cardiac cost, the threshold is set at 40%, marking the onset of physiological overload and acute cardiovascular risk during prolonged exertion in heat [5, 10]. For skin conductance level, we use $20\mu\text{S}$, above which sympathetic arousal transitions from moderate stress to extreme activation [1, 6, 7].

2.2 Participants

We recruited 11 rickshaw puller participants (P1–P11) from Dhaka (Table 2) through convenience sampling. Participants ranged in age from 33 to 70 years (mean 56.2) and were all male and married. Six participants had attended schools up to grade 8, while five reported no formal education. Daily income ranged from USD 3 to USD 7, and reported work hours ranged from 7 to 14 hours per day, with work windows spanning as early as 7 AM and as late as midnight. Among the participants, 8 of 11 (73%) reported body pain, 3 (27%) reported cardiovascular disease, and 1 (9%) reported asthma (Table 2).

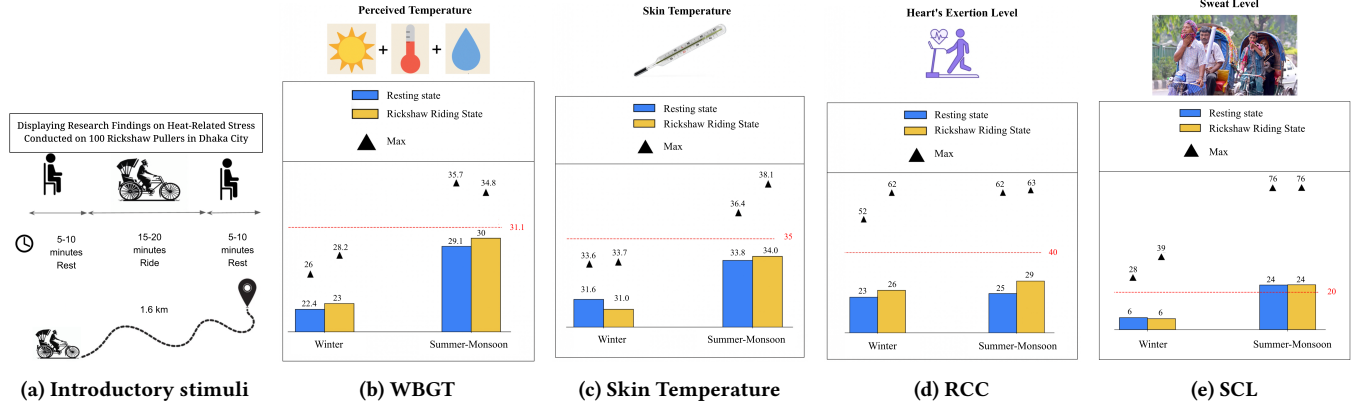
2.3 Study Protocol

Each participant went through two phases. The first phase replicated the sensor data collection procedure, which we followed for the cohort of 100 rickshaw pullers (see [11] for details). This ensures that the interview participants get familiarized with what that cohort of 100 rickshaw pullers had gone through. Both phases were conducted in the participants’ local language, Bangla.

During second phase, the interviewer first oriented the participant to the activity using the rest-ride-rest schematic (Figure 2a).

Table 1: Comparison of personal informatics studies with low-literacy or frontline worker populations.

Study	Subject Group	Devices	Subject Literacy	Data Window	Presentation
Liu et al., 2025 [8]	General population	—	—	80 years (1940–2020)	Aggregated
Solano-Kamaiko et al., 2025 [13]	Home-care workers	Fitbit, Oura	High School –Master’s	4 weeks (n=20)	Individual
Okolo et al., 2024 [9]	CHWs	iPad	Middle School –Master’s	Just-in-time (n=35)	Patient image
Chaudhary et al., 2025 [4]	Farmers	Polar HR + tempi.fi	Middle School	2 weeks (n=3)	Individual
Ours	Rickshaw pullers	Embrace Plus + Temp & Hum	No formal–Elementary	3 years (2023–2025)	Aggregated

**Figure 2: Visual stimuli including the physiological measurements across winter and summer-monsoon seasons, showing resting state (blue) and rickshaw riding state (yellow). Red dashed lines indicate clinically validated occupational thresholds.**

For each biomarker, the interviewer (1) defined and explained the concept and its risk threshold in plain Bangla, then (2) asked the participant to rate the perceived risk of that biomarker on a five-point scale, anchored by a fan speed-regulator analogy where 1 denotes no risk, and 5 denotes extreme risk. The interviewer (3) then presented the corresponding visualization, walking through the rest/ride and seasonal comparisons and the threshold crossings, and (4) asked the participant for a second rating on the same scale.

Finally, (5) for each of the four biomarkers, the interviewer used a common set of open-ended, semi-structured questions. The questions consist of whether the sensation (heat, skin temperature, cardiac exertion, or sweating) felt tolerable during work and, if not, when and why; which season it felt most severe in; whether it was felt more during rest or while riding; how long into a ride the sensation typically began and how long it persisted; what the participant did in response; and whether those measures felt sufficient.

Each interview lasted approximately 24 minutes ($SD \pm 5$ minutes). All experimental protocols were reviewed and approved by the authors' Institutional Review Board (IRB) in accordance with relevant ethical guidelines and regulations. Interviews were conducted by one of three interviewers who are also the authors, with at least two interviewers typically present during the session. Prior to participation, each participant was briefed on the study objectives, their right to withdraw at any time without consequence, and the expected duration of their involvement. Oral informed consent was obtained (considering participants' lack of formal education) from all participants before data collection began. In addition to standard fare for the rickshaw trip, participants received separate compensation for their time participating in the interview.

2.4 Analysis

We record each participant's pre-exposure rating, post-exposure rating, their difference, and relevant descriptive statistics (e.g., median, IQR, mean, and standard deviation) for all four biomarkers, allowing us to examine whether viewing the visualizations shifted perceived risk and in which direction. Besides, to assess whether pre/post risk ratings differed for each biomarker, we conducted one-sided Wilcoxon signed-rank tests, testing the hypothesis that post-visualization ratings were stochastically greater than pre-visualization ratings. This non-parametric test was chosen given the ordinal 1–5 rating scale and small sample size. Tests were run per biomarker on paired participant ratings.

Then, three authors perform Thematic Analysis on the translated English transcripts, following the six steps described by Braun and Clarke in Thematic Analysis [2]. First, we apply semantic coding, i.e., codes that directly reflect the participants' observations, keeping the analysis grounded in their accounts. Each interview is stored as a separate docx structured as a sequence of interviewer–rickshaw puller exchanges, and during coding, we highlight each relevant segment and record its code inline beside the corresponding line. We maintain a codebook as a spreadsheet in which rows correspond to participants and columns to codes, marking the presence or absence of each code for each interview. The three authors then met to examine patterns of similarity across the codebook and extract distinctive, representative themes. This process yielded 28 codes and 4 themes, which we present in the following subsection.

3 Findings

Figure 3 presents pre and post-visualization risk ratings across all four biomarkers for the ten participants with complete data (P1 excluded). SCL and RCC received consistently high pre-visualization ratings, reflecting participants' immediate bodily familiarity with

Table 2: Demographic Profile of Participants

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11
Age	60	70	65	33	65	45	45	54	45	69	67
Education	Gr.1	Gr.8	Gr.3	Gr.4	NA	Gr.1	Gr.5	NA	NA	NA	NA
Marital status	M	M	M	M	M	M	M	M	M	M	M
Income/day (USD)	5–6	5–6	5–6	6	4–5	5–6	6–7	4–5	5–6	3–4	7
Work hours/day	7	13	8	10–11	11	7	11	10	9	9	14
Work window	8–15	8–24	7–15	9–22	11–22	8–15	9–20	12–22	9–18	8–17	7–21
Meals/day	3	2	2	3	2	3	2	3	3	3	3
Sleep hours/day	6	4	4	8	3–4	8	7	7–8	8	5–6	6
Smoking	✓	×	✓	×	✓	×	×	×	✓	×	×
CVD	×	✓	×	×	✓	×	×	×	×	✓	×
Asthma	×	×	×	×	×	×	×	×	×	✓	×
Body Pain	✓	✓	×	✓	✓	✓	×	✓	✓	✓	×

sweating and cardiac strain. WBGT and skin temperature showed greater variability, with several participants initially assigning moderate ratings. Table 3 reports descriptive statistics and one-sided Wilcoxon signed-rank results for each biomarker. Variance narrowed after visualization for WBGT, skin temperature, and especially RCC (SD dropping from 1.0 to 0.6), suggesting the visualization brought participants toward a shared assessment rather than simply raising ratings on average. SCL was the exception on every measure: already at the ceiling before visualization (median 5), it showed negligible mean shift, a slight increase in variance, and the weakest Wilcoxon signal ($p = .564$), consistent with Theme 2's finding that sweat is the most immediately self-perceptible biomarker and therefore the least informative to visualize. None of the four shifts reached statistical significance ($p < .05$), considering the small sample size; RCC came closest ($W = 0.0$, $p = .059$), aligning with its largest mean shift and greatest variance reduction.

In the subsequent sections, we present the qualitative findings from our thematic analysis alongside the quantitative risk rating outcomes from Figure 3.

3.1 Theme 1: Perception of Extreme Heat Relies on Multiple Educational Factors

Rickshaw pullers in Dhaka endure extreme ambient heat for 8–10 hours a day, making them among the city's most heat-vulnerable outdoor workers.

Knowledge of external heat sources. Beyond ambient heat, participants identified radiant heat from paved roads, engine exhaust, and direct solar radiation as compounding stressors, and several highlighted the rickshaw itself, whose metal frame, seat, and handlebars grow too hot to handle. P9 noted: “Sometimes the handle of the rickshaw, the seat and the paddle get so heated that you can't sit in the rickshaw anymore.” Traffic congestion was a particularly acute amplifier; P3 captured the combined effect: “When I get stuck in traffic, there is heat from the vehicle engines, heat rising from the paved road, and heat from above. At that time, only by God's protection do we survive.”

Health knowledge: intuitive awareness alongside misinformation. Despite little or no formal education, participants held practical health knowledge accumulated through fellow workers and passengers. P6 articulated a clear strategy: “After resting, drinking water, eating biscuits, and sitting in the shade, I feel better.” P8 showed awareness of cardiac vulnerability, noting that “since we work so hard, we don't even realize when heart problems develop.” Yet this

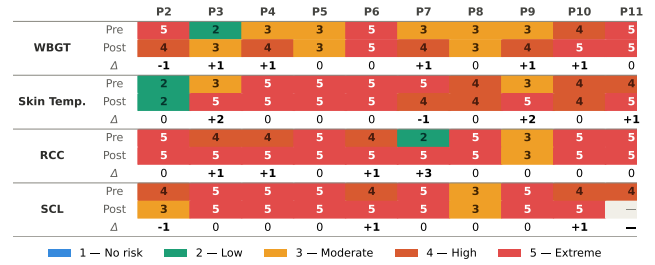


Figure 3: Risk ratings (1–5) assigned by each participant. Scale: 1=No Risk, 2=Low, 3=Moderate, 4=High, 5=Extreme.

Table 3: Descriptive statistics of risk ratings.

Biomarker	n	Pre		Post		Mean Δ	Wilcoxon-Test W	p
		Median (IQR)	Mean (SD)	Median (IQR)	Mean (SD)			
WBGT	10	3 (3–4.8)	3.6 (1.1)	4 (3.2–4.8)	4.0 (0.8)	0.4	3.5	0.103
Skin Temp.	10	4 (3.2–5)	4.0 (1.1)	5 (4–5)	4.4 (0.9)	0.4	1.5	0.194
RCC	10	4.5 (4–5)	4.2 (1.0)	5 (5–5)	4.8 (0.6)	0.6	0.0	0.059
SCL	9	5 (4–5)	4.4 (0.7)	5 (5–5)	4.6 (0.9)	0.1	2.0	0.564

knowledge was inconsistently reliable: P2 had internalized a misapplied instruction, “It is not right to drink saline (ORS) in a healthy body,” while P4 read exertion-induced palpitations as gastric discomfort, self-medicating with antacids.

3.2 Theme 2: Risk Factors Beyond Biomarkers

Beyond the sensor-based markers, interviews surfaced a richer picture of occupational risk that our sensing measures do not capture.

Frequency and timing of extreme heat. Participants universally agreed that heat risk concentrates in summer and monsoon, with winter posing minimal danger, though estimates of extremely hot days per month varied widely, from 7–8 days (P3) to a whole month (P9). This likely reflects differences in individual adaptation thresholds rather than disagreement about conditions. They associated the worst heat with specific Bangla-calendar months, spanning Boishakh through Bhadra (April–September), i.e., pre-monsoon through early autumn. Within a day, a consistent peak window ran from roughly 10 AM to 3 PM; at the trip level, onset of significant discomfort during a 20-minute ride was reported variably, between 3 and 10 minutes.

Passenger interaction as an additional risk factor. A risk not captured by any biomarker is the physical and psychological pressure of passenger interaction. Demands for faster speeds introduce both exertion and emotional stress, elevating cardiac load and hastening heat-related deterioration. P3 described the effect: “If someone says ‘pull a little harder,’ and I pull the rickshaw a bit harder, then my heart starts beating fast.” P7 noted that carrying two passengers made sweat drying impossible. These interactions are unpredictable and fall outside the puller's control, a structural source of risk any intervention must acknowledge.

3.3 Theme 3: Recovery Approaches Vary by Biomarker and Economic Conditions

Participants' ability to recover from heat-related strain was shaped by two intersecting forces: the recovery strategies available and the economic constraints.

Natural and self-managed recovery. Participants relied primarily on natural, low-cost interventions, favored precisely because they require no expenditure: resting in shade, sitting under trees, and catching natural wind were the most common first-line responses.

Beyond passive rest, they described active routines, drinking water or lemonade, wiping the body with a wet towel, eating biscuits, and consuming oral rehydration salts (ORS). P4 described a typical mid-trip sequence: *“If it feels too hot, I pull over and rest. I drink water, eat something... If I drink lemonade and saline, it helps relieve body weakness.”* Seasonal intensity shaped perceived necessity; P9 found electrolyte balance obligatory in the hottest months: *“During summer, I have to consume saline to stay healthy.”*

Economic barriers to full recovery. Despite knowing what would help, participants frequently could not access better recovery resources for financial reasons. Nutritious options, e.g., coconut water, pomegranate and orange juice, milk, were consistently identified as desirable but unaffordable. P8 was direct about the structural reality: *“Eating well is a benefit, but we don’t get those things. We sweat and sleep anyway.”* The pressure to maximize earnings within limited hours further undermines rest, producing a chronic tension between physical need and financial necessity. P3 articulated the trade-off plainly: *“The risk doesn’t go away. Even if I think I would wait to fully recover, I won’t get any work done. So even if there’s a little problem in the body, I go back to work.”* P3 described how a short trip expands into sustained exertion: *“If I take a trip to Mohammadpur (a destination), it’s not just twenty minutes. If there is traffic, it can take an hour... I cannot rest, because the trip can extend for hours.”* Even brief rest offers only transient relief (P6: the body *“cools a little, but later it heats up again”*).

3.4 Theme 4: Normalizing Occupational Risk

Across participants, normalization operated through physiological adaptation and cognitive-economic normalization.

Physiological heat adaptation. A subset of participants described robust heat tolerance, attributing their endurance to long occupational exposure. P3, who worked through midday without reported exhaustion, framed this in terms of providence: *“By Allah’s mercy, I can work a lot during noon... after one trip I get another, and even with that effort, I don’t get exhausted.”* For others, tolerance had been built incrementally; P2, who found extreme heat uncomfortable but not unbearable, had developed a consistent post-exertion routine of wiping down and washing with water when available.

Normalization through limited awareness. For several participants, heat had been so absorbed into the texture of daily work that it no longer registered as a discrete hazard. P1 rated his own RCC risk as only moderate because he had no known heart condition: *“It would have been a problem if there had been damage”*, treating the absence of identifiable harm as proof of safety, a pattern that discounts chronic risk. P8 reduced seasonal variation to unremarkable routine: *“It feels hot in the summer, cold in the winter, that’s it.”* In both cases, normalization served as a cognitive shortcut that made continued work psychologically sustainable while marginalizing accurate risk appraisal.

Normalization through economic necessity. For others, normalization was less a perceptual failure than a rational adaptation to circumstances offering no alternative. P2 stated the logic plainly: *“Even if you can’t stand it, you have to endure it.”* P5 came closest to naming the constraint, trailing off mid-explanation: *“I want to (take a break). But even if I want to —”* The unfinished sentence is itself revealing: the barrier to rest is not motivation but the conditions of informal labor that make rest economically punishing.

4 Discussion

We organize this discussion around what our findings show, whether they can be trusted given our study design, why the visualizations were effective, what these findings suggest for future systems, and how far the results generalize.

4.1 Interaction with Visualization

Visualization produced a net positive shift in risk ratings (Figure 3). Among the markers, SCL’s low average delta is consistent with it being the most immediately and viscerally perceptible biomarker: participants already rated it high before seeing any data, leaving little room for upward revision. RCC’s large delta reflects the opposite dynamic: cardiac strain during exertion is harder to self-monitor precisely, making the aggregate data genuinely informative. RCC also showed the most consistent directional shift among participants (Wilcoxon signed-rank, $W = 0.0$, $p = .059$, Table 3), the closest of the four biomarkers to conventional significance, suggesting cardiac strain is one of the biomarkers for which visualization is most impactful.

Together, these patterns support a broader design principle: visualizations of physiological risk are most impactful for indicators that are real but not easily self-perceived, and least impactful where the risk is already legible through direct physical experience.

4.2 Interviewer Effects on Visualization

These risk-rating shifts, however, cannot be attributed solely to the visualization. Our study does not include a control condition, and our design cannot fully separate the effect of the visualization itself from that of the interviewer’s verbal explanation that accompanied it. The reported shifts reflect the combined effect of both. We treat visualization-plus-explanation as the intervention unit under this study, rather than claiming to isolate the visualization’s independent contribution. Besides, three interviewers conducted the sessions, which introduces a further source of variability beyond the visualization itself. This limitation reflects a structural trade-off common to work with low-literacy, frontline worker populations [4, 8, 9].

4.3 Visualization Explainability

A key finding of this study is that well-designed, low-text visualizations of wearable data can communicate physiological risk to participants with little or no formal education, provided the visual grammar is explained at the outset. Even participants who could not read independently were able to interpret bar heights, threshold lines, and iconographic cues once the interviewer introduced the legend. The pre- and post-ratings in Figure 3 provide direct evidence: shifts occurred across all four biomarkers. These results are consistent with prior work cautioning against complex numerical labels and multi-layered color schemes for low-literacy users [9].

However, comprehension of the visualization varied clearly across visual elements. The rest-ride-rest schematic and the blue/yellow color coding were understood immediately, with no participant requiring repeated explanation. The threshold line and maximum-value marker were also understood, with several participants correctly identifying when a value fell above or below the danger line without needing repeated clarification. Some biomarker icons were less immediately legible, e.g., the icon for cardiac exertion required

the interviewer to add a concrete analogy, comparing rickshaw pulling to running on a treadmill, before participants interpreted it correctly.

4.4 Potential Design Considerations

Our analysis surfaces potential design considerations in four different aspects.

Just-in-Time Interventions. The irregular, passenger-driven structure of rickshaw work means that mid-ride alerts are largely impractical, as pullers cannot stop while a passenger is aboard, and income pressure means even empty-rickshaw breaks are perceived as costly. The more viable window is the natural pause between rides, when a puller has just dropped a passenger and is momentarily free. A just-in-time system could use this window to prompt one of three categories of recovery action that participants already practice: hydration and nourishment (water, ORS, lemon water, or food such as biscuits or fruit), rest and cooling (shade-seeking, sitting under trees, or slow movement to catch wind), or hygiene (wiping sweat). Because these behaviors are already part of pullers' daily routine, the intervention need not introduce new habits. Considering financial constraints, prompting a puller to purchase juice or take a 30-minute break without acknowledging affordability or scheduling uncertainty will be ignored or resented. Infrastructure-level solutions, e.g., public water points, shaded rest stops near busy intersections, are more equitable entry points than personal device recommendations that presuppose smartphone ownership or discretionary spending.

Post-Hoc Reflection. Our visualization sessions demonstrate that post-hoc reflection on aggregate physiological data can shift risk perception meaningfully, even in participants with no formal education and no prior exposure to biomarker concepts (Figure 3, for RCC, $\bar{\Delta} = 0.6$). This form of intervention does not require real-time sensing or connectivity. A periodic session (monthly or yearly e.g., early summer) in which a community health worker presents cohort-level data using the low-literacy visualization format we developed could serve as a scalable awareness tool. In Bangladesh, where rickshaw pullers represent a significant portion of the urban informal workforce, such sessions could be integrated into existing government occupational health policy. Our visualization format requires no literacy, no personal device, and no ongoing infrastructure, making it a low-cost policy-level intervention.

Implications for Interface Design. Beyond intervention timing, our findings point to concrete interface design choices. First, icon-based, binary visuals were interpretable by participants who could not read independently. This pattern extends beyond low-literacy contexts: binary visuals are also effective for general audiences [8]. Second, a threshold-anchored reference point per marker was likewise sufficient for participants to identify risk, arguing for minimal, single-metric encodings over multi-value dashboards. Third, we grounded an abstract severity scale in a familiar physical analogy, a fan speed regulator, suggesting that unfamiliar biometric concepts may transfer better through familiar, everyday referents than through numerical or Likert-style scales. Finally, participants needed an interviewer to introduce this visual grammar before they could interpret it unaided. Future interfaces should budget for an

onboarding step rather than assume users can interpret visuals without guidance.

Implications for Wearable Computing. This onboarding requirement points to a broader implication: personal informatics research has largely assumed a model in which the wearer directly views their own data on a device. Our findings instead suggest a model in which wearable-derived value is delivered later, in aggregate, and through a human intermediary, without requiring the end user to own or operate a device, useful for populations facing literacy, connectivity, or device-ownership constraints. Furthermore, our study offers visualizations and thresholds for a population largely absent from wearable sensing and ubiquitous computing literature.

4.5 Generalizability

We expect these findings to generalize along two axes. First, by design approach: the effect should extend beyond heat stress to any domain where workers face measurable environmental risk, literacy is limited, and individual data is impractical, making aggregate, threshold-anchored, icon-supported visualizations a broadly applicable strategy. Second, by occupation: the coping strategies we document, e.g., hydration and shade-seeking, are shared by workers in structurally similar conditions, such as construction, farming, and gig delivery, who face the same interplay of exertion, financial pressure, and irregular schedules.

5 Limitations

This study has several limitations. Our sample of 11 participants, while sufficient for qualitative thematic analysis, limits the generalizability of the quantitative risk rating shifts. Besides, participants were recruited from a single urban area in Dhaka, which may not reflect the experiences of rickshaw pullers in smaller cities or rural settings. Additionally, interviews were conducted in a single session, meaning the long-term effect of visualization on behavior change, as opposed to immediate perception shift, remains unknown. Most importantly, we treat the study as exploratory and hypothesis-generating rather than confirmatory. The qualitative themes are the study's primary contribution, reflecting thematic outcomes across a homogeneous participant group, while the quantitative shifts are observed signals that motivate future work.

6 Conclusion

Rickshaw pullers in Dhaka can meaningfully interpret aggregated physiological risk data with some guidance, and exposure to such data is associated with a shift in reported risk perception, though this shift did not reach statistical significance at our sample size. Our thematic analysis shows that normalization, financial constraints, and the structural unpredictability of participants' informal labor mediate this perception. Further, we discuss the possibility of three design considerations (just-in-time, post-hoc, and designing interface) for low-literacy outdoor workers.

Going forward, there is scope for future work to formalize these observations by asking participants to rank or score each visual element individually, isolating which components most reliably transfer information. Besides, future work with a larger sample size can examine whether the effects described in this study hold up over repeated or longitudinal sessions.

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